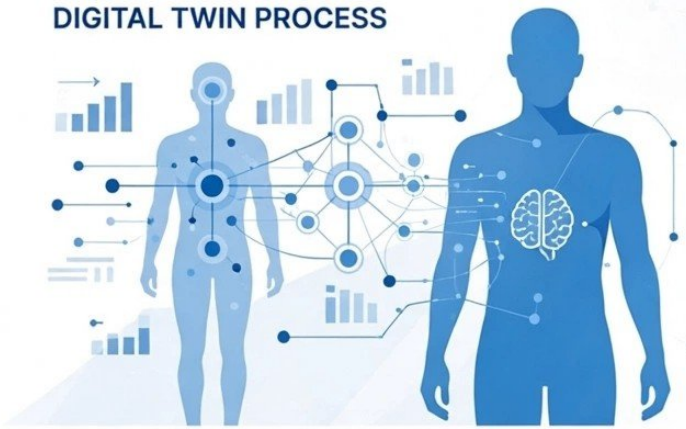
GlucoTwin — Digital Twin for Type 2 Diabetes

Team Death Note | IIT Jodhpur

Team Leader:
Yug Purush Balyan

Member:
Shikha



Real-Time Patient Data → AI-Powered Modeling → Predictive Insights & Care

Happiest Health Reimagining Healthcare Summit 2026

Hackathon 2026 • Healthcare Innovation • Personalized Predictive Care • IIT Jodhpur

SLIDE 02 • PROBLEM STATEMENT

Why India Needs GlucoTwin?

- India has **100M+ Type 2 Diabetes** patients
- Current care is **REACTIVE** not proactive
- No fusion of **EHR + Wearable** data in clinics

Solution: Proactive Digital Twin predicts glucose spike 2 hours in advance.

Use Case: Hyperglycemia Prediction

Technical Implementation

Impact & Tech Stack

Datasets

- **Synthea Synthetic EHR** – simulated electronic health records for patient journeys and demographics
- **OhioT1DM Wearable open dataset** Real-world CGM & wearable data for Type-1 diabetes

Model

1 hour data predicts next 2 hours glucose

Risk

Low • Stable glucose trend

Med • Approaching threshold

High Hypoglycemia risk

Tech Stack

Python
data processing & training

LSTM
sequence model for time-series prediction

FastAPI
Inference API backend

Streamlit
Interactive web dashboard

Plotly **Plotly**
dynamic glucose trend visualizations

What-If Simulator

Interactive scenario testing: adjust meals, insulin, activity to forecast glucose and evaluate risk outcomes in real time

GlucoTwin Dashboard - Live Digital Twin

Real-time monitoring • Patient digital twin • Live telemetry

● LIVE • Connected • Last updated: 2 seconds ago

Virtual Patient



54
Male

Patient ID: VT-1024
Age: 54 • Male • BMI: 26.3

Status: Stable ✓ Stable

Weight: **82.4 kg**

Height: **178 cm**

Diabetes Type: **Type 2**

Digital Twin Model
Accuracy: 96.2%
Updated: Oct 15, 2024 09:42

Live Graphs

Glucose Level (24h) — mg/dL



Heart Rate (bpm)



78 bpm

Glucose	Heart Rate
112 mg/dL	78 bpm
↓ -5 mg/dL • Trend: Decreasing	Normal • 65-95 bpm

Alert System

Alerts: 2 Active

Moderate hypoglycemia risk
Predicted within 2h • Confidence 74% •
Recommendation: Monitor carbs intake

Irregular heart rate detected
Occasional PVCs observed •
Review ECG recommended

No critical events
All other vitals within normal range

Alert System: Enabled

Time In Range (24h): 84%

Risk Score: Low-Medium

Data Feed: Secure • HIPAA Compliant

Impact & Future

Impact & Future Roadmap

Impact

- Reduces emergency hyperglycemia by 30%
- Saves doctor time
- Privacy compliant

Future

- Insulin recommendation engine
- Google Fit integration
- Expand to Hypertension Digital Twin

Thank You